

Homework 3

Andrew Hwang

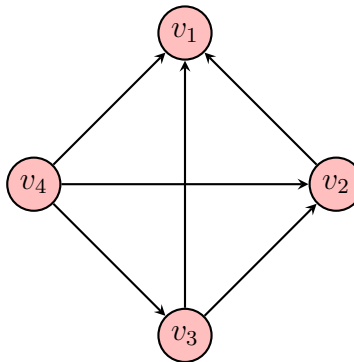
Math 38

Gayee Park

April 25, 2026

Question 1. Prove or disprove for each $n \geq 1$: Every simple digraph with n vertices has two vertices with the same out-degree or two vertices with the same in-degree.

Proof. We shall disprove by counterexample as shown by the graph below:



In this graph, v_1 has in-degree 3 and out-degree 0, v_2 has in-degree 2 and out-degree 1, v_3 has in-degree 1 and out-degree 2, and v_4 has in-degree 0 and out-degree 3. Hence, no two vertices have the same in-degree nor out-degree, disproving this statement. $\square$

Question 2. Let $d_1, \dots, d_n$ be a list of positive integers with $n \geq 2$. Prove that there exists a tree with degrees equal to such d_i 's if and only if $\sum d_i = 2n - 2$.

Proof. ($\implies$) In the forward direction, using the handshake lemma,

$$\sum_i^n d_i = 2|E|.$$

Since all trees have $n - 1$ edges,

$$\sum_i^n d_i = 2(n - 1) = \boxed{2n - 2}.$$

($\impliedby$) In the reverse direction, let us use induction by inducting on n . Trivially, the base case $n = 2$ results in the degree sequence $(1, 1)$ because the degrees must be positive. Now assume the statement holds for all lists of $k - 1$ positive integers. Suppose we are given a list $d_1, \dots, d_k$ of positive integers with $\sum_{i=1}^k d_i = 2k - 2$. Since every tree with at least two vertices has a leaf, we need at least one $d_i = 1$. Without loss of generality, suppose $d_k = 1$.

Since $d_k = 1$, a leaf must be attached to some other vertex. Without loss of generality, suppose it is attached to the vertex of degree d_1 . Remove this leaf and reduce d_1 by 1 to obtain the list $d_1 - 1, d_2, \dots, d_{k-1}$. It must be true that $d_1 - 1 \geq 1$ since if $d_1 = 1$ then $\sum d_i \leq k < 2k - 2$ for $k \geq 3$, which is a contradiction. This new list has $k - 1$ entries, all positive integers and its sum is

$$(d_1 - 1) + d_2 + \dots + d_{k-1} = (2k - 2) - 1 - 1 = 2(k - 1) - 2.$$

By the induction hypothesis, a tree T^* with $k - 1$ vertices realizes this degree sequence. Adding a new leaf vertex v_k and connecting it to the degree vertex $d_1 - 1$ forms a tree T on k vertices realizing $d_1, \dots, d_k$. This completes the induction. $\square$

Question 3. Let T be a tree in which every vertex has either degree 1 or degree k . Determine the possible values of the number of vertices of this T .

Proof. If T has a vertex with degree 1, we define that as a leaf. Hence, let us denote ℓ as the number of leaves, and m as the number of non-leaves. Hence, $n = \ell + m$. Let us use the proof from Question 2. Then, we know that

$$\sum d_i = 1(\ell) + k(m).$$

However, from problem (2), we also know that

$$\sum d_i = 2n - 2 = 2(\ell + m) - 2.$$

Setting those two equal gives

$$\ell + km = 2(\ell + m) - 2 \implies \ell = (k - 2)m + 2.$$

Hence,

$$n = \ell + m = (k - 2)m + 2 + m = (k - 1)m + 2.$$

and we are given the set of possible values of vertices N , where

$$N = \{(k - 1)m + 2, \quad k, m \in \mathbb{N}\}$$

□

Question 4. Use Cayley's Formula to prove that the graph obtained from a complete graph K_n by deleting an edge has $(n - 2)n^{n-3}$ spanning trees.

Proof. By Cayley's Formula, K_n has n^{n-2} spanning trees. Let e be the edge to be deleted. We count how many spanning trees of K_n contain e . Each spanning tree has exactly $n - 1$ edges, and K_n has $\binom{n}{2} = \frac{n(n-1)}{2}$ edges. By symmetry, every edge of K_n appears in the same number of spanning trees. Hence the total number of trees containing e is

$$= \frac{(n - 1) \cdot n^{n-2}}{\frac{n(n-1)}{2}} = 2n^{n-3}.$$

A spanning tree of $K_n - e$ is precisely a spanning tree of K_n that *does not* use e . Therefore, by principles of counting, the number of spanning trees of $K_n - e$ is

$$n^{n-2} - 2n^{n-3} = n^{n-3}(n - 2) = \boxed{(n - 2)n^{n-3}}.$$

□

Question 5. Prove that an edge e of a connected graph G is a cut-edge if and only if e belongs to every spanning tree. Prove that e is a loop if and only if e belongs to no spanning tree.

Proof. We split this into two parts.

Part 1: e is a cut-edge if and only if e belongs to every spanning tree.

($\implies$) Suppose $e = \{u, v\}$ is a cut-edge of G . Then $G - e$ is disconnected. Any spanning tree T of G must be connected and span all vertices. If T did not contain e , then T would be a connected spanning subgraph of $G - e$, causing a contradiction.

($\impliedby$) Suppose e belongs to every spanning tree. Assume for contradiction that e is not a cut-edge, so $G - e$ is connected. Then $G - e$ contains a spanning tree T . But T is also a spanning tree of G that does not contain e , contradicting the assumption that e belongs to every spanning tree. Therefore e is a cut-edge.

Part 2: e is a loop if and only if e belongs to no spanning tree.

($\implies$) Suppose e is a loop at vertex v . Since a loop is a cycle of length one, no spanning tree can contain e because trees are acyclic. Hence, e belongs to no spanning tree.

($\impliedby$) Suppose e belongs to no spanning tree. Assume for contradiction that e is not a loop, so $e = \{u, v\}$ with $u \neq v$. Since G is connected, it must have a spanning tree T . If $e \in T$, the statement trivially contradicts. If $e \notin T$, then adding e to T creates exactly one cycle C . Removing any other edge $e' \neq e$ of C from $T + e$ yields a new spanning tree $T^* = T + e - e'$ that contains e , which contradicts again. Therefore, e must be a loop. $\square$